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need not indicates permission not to do something

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will indicates that something is certain or expected to happen as a result of action taken by an agency the behaviour of which is outside the scope of the present document

will not indicates that something is certain or expected not to happen as a result of action taken by an agency the behaviour of which is outside the scope of the present document

might indicates a likelihood that something will happen as a result of action taken by some agency the behaviour of which is outside the scope of the present document

might not indicates a likelihood that something will not happen as a result of action taken by some agency the behaviour of which is outside the scope of the present document

In addition:

is (or any other verb in the indicative mood) indicates a statement of fact

is not (or any other negative verb in the indicative mood) indicates a statement of fact

The constructions "is" and "is not" do not indicate requirements.

1 Scope

The present document specifies the stage 3 protocol and data model for the Npanf Service Based Interface. It provides stage 3 protocol definitions and message flows, and specifies the API for each service offered by the 5G PAnF.

The 5G System stage 2 architecture and procedures are specified in TS 23.501 [2] and TS 23.502 [3].

The Technical Realization of the Service Based Architecture and the Principles and Guidelines for Services Definition are specified in TS 29.500 [4] and TS 29.501 [5].

2 References

The following documents contain provisions which, through reference in this text, constitute provisions of the present document.

- References are either specific (identified by date of publication, edition number, version number, etc.) or non-specific.
- For a specific reference, subsequent revisions do not apply.
- For a non-specific reference, the latest version applies. In the case of a reference to a 3GPP document (including a GSM document), a non-specific reference implicitly refers to the latest version of that document *in the same Release as the present document*.

[1] 3GPP TR 21.905: "Vocabulary for 3GPP Specifications".

[2] 3GPP TS 23.501: "System Architecture for the 5G System; Stage 2".

[3] 3GPP TS 23.502: "Procedures for the 5G System; Stage 2".

[4] 3GPP TS 29.500: "5G System; Technical Realization of Service Based Architecture; Stage 3".

[5] 3GPP TS 29.501: "5G System; Principles and Guidelines for Services Definition; Stage 3".

[6] OpenAPI: "OpenAPI Specification Version 3.0.0", https://spec.openapis.org/oas/v3.0.0.

[7] 3GPP TR 21.900: "Technical Specification Group working methods".

[8] 3GPP TS 33.501: "Security architecture and procedures for 5G system".

[9] IETF RFC 6749: "The OAuth 2.0 Authorization Framework".

[10] 3GPP TS 29.510: "5G System; Network Function Repository Services; Stage 3".

[11] IETF RFC 7540: "Hypertext Transfer Protocol Version 2 (HTTP/2)".

[12] IETF RFC 8259: "The JavaScript Object Notation (JSON) Data Interchange Format".

[13] IETF RFC 7807: "Problem Details for HTTP APIs".

[14] 3GPP TS 33.503: "Security Aspects of Proximity based Services (ProSe) in the 5G System (5GS)".

[15] 3GPP TS 29.571: "5G System; Common Data Types for Service Based Interfaces; Stage 3".

3 Definitions and abbreviations

3.1 Definitions

For the purposes of the present document, the terms and definitions given in 3GPP TR 21.905 [1] and the following apply. A term defined in the present document takes precedence over the definition of the same term, if any, in 3GPP TR 21.905 [1].

3.2 Abbreviations

For the purposes of the present document, the abbreviations given in 3GPP TR 21.905 [1] and the following apply. An abbreviation defined in the present document takes precedence over the definition of the same abbreviation, if any, in 3GPP TR 21.905 [1].

PAnF Prose Anchor Function

4 Overview

The Prose Anchor Function (PAnF) is the network entity in the 5G Core Network (5GC) supporting security procedure over Control Plane for the 5G ProSe UE-to-Network relay as specified in 3GPP TS 33.503 [14]. Within the 5GC, the PAnF offers services to the AUSF, SMF via the Npanf service based interface.

Figure 4-1 provides the reference model (in service based interface representation and in reference point representation), with focus on the PAnF:

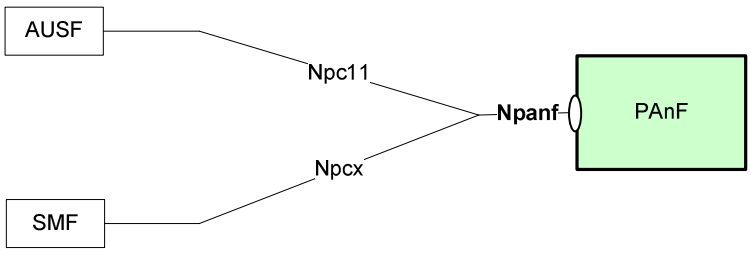

Figure 4-1: Reference model – PAnF

Editor's note: The reference point between PAnF and SMF will be aligned with Stage 2.

The PAnF supports the following functionalities:

- Storage and retrieval of ProSe context information of the 5G ProSe Remote UE;
- Checking whether the 5G ProSe Remote UE is authorized to use the UE-to-Network Relay service.
- Resolving the ProSe Remote User ID to SUPI.

5 Services offered by the 5G PAnF

5.1 Introduction

Table 5.1-1 shows the PAnF Services and PAnF Service Operations:

Table 5.1-1: List of 5G PAnF Services

Service Name	Service Operations	Operation Semantics	Example Consumer(s)
Npanf_ProseKey	Register	Request/Response	AUSF
	Retrieve	Request/Response	AUSF
Npanf_ResolveRemoteUserId	Get	Request/Response	SMF

Table 5.1-2 summarizes the corresponding APIs defined for this specification.

Table 5.1-2: API Descriptions

Service Name	Clause	Description	OpenAPI Specification File	apiName	Annex
Npanf_ProseKey	6.1	PAnF Prose Key Service	TS29553_Npanf_ProseKey.yaml	npanf-prosekey	A.2
Npanf_ResolveRemoteUserId	6.2	Resolve Remote User ID Service	TS29553_Npanf_ResolveRemoteUserId.yaml	npanf-userid	A.3

5.2 Npanf_ProseKey Service

5.2.1 Service Description

The Npanf_ProseKey service enables an NF to request the PAnF to store the ProSe context information for a 5G ProSe Remote UE or to request CP-PRUK from the PAnF. The following are the key functionalities of this NF service.

- Store the Prose context information;
- Retrieve the Prose Key.

5.2.2 Service Operations

5.2.2.1 Introduction

The service operations defined for the Npanf_ProseKey service are as follows:

- Register: It allows a consumer NF to store the Prose Context information;
- Retrieve: It provides to the NF service consumer of the CP-PRUK matching input criteria.

5.2.2.2 Register

5.2.2.2.1 General

The Register service operation is invoked by a NF Service Consumer to request the PAnF to store the ProSe context information for a 5G ProSe Remote UE.

The Register service operation is used during the following procedure:

- PC5 security establishment for 5G ProSe UE-to-Network relay communication over Control Plane (see 3GPP TS 33.503 [14], clause 6.3.3.3.2)

The NF Service Consumer (i.e. AUSF) shall request the PAnF to store the ProSe context information for a 5G ProSe Remote UE as shown in Figure 5.2.2.2.1-1

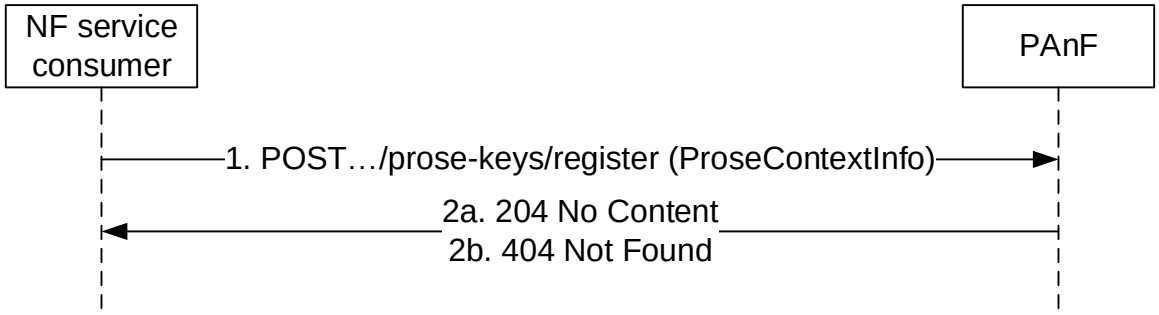

Figure 5.2.2.2.1-1: Prose Keys registration

1. The NF service consumer (e.g. AUSF) sends a POST request to the resource representing the Prose Context Info for a 5G ProSe Remote UE.

2a. On success, the PAnF responds with "204 No Content".

2b. If the user does not exist, HTTP status code "404 Not Found" shall be returned including additional error information in the response body (in the "ProblemDetails" element).

On failure, the appropriate HTTP status code indicating the error shall be returned and appropriate additional error information should be returned in the POST response body.

5.2.2.3 Retrieve

5.2.2.3.1 General

Figure 5.2.2.3.1-1 shows a scenario where the NF service consumer (e.g. AUSF) sends a request to the PAnF to retrieve the CP-PRUK. The request contains the CP-PRUK ID and Relay Service Code.

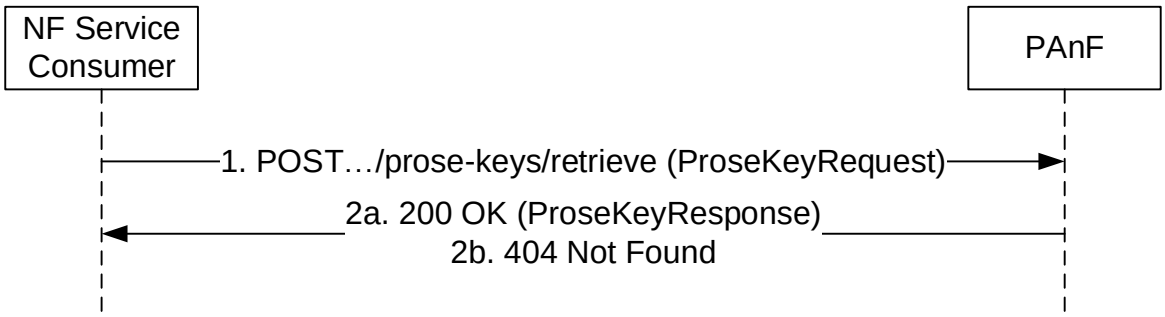

Figure 5.2.2.3.1-1: Prose Keys retrieval

1. The NF service consumer (e.g. AUSF) sends a POST request to the resource representing the Prose Key.

2a. On success, the PAnF responds with "200 OK" with the message body containing the ProseKeyResponse.

2b. If the user does not exist, or the ProSe Key does not exist in the in the PAnF, HTTP status code "404 Not Found" shall be returned and additional error information should be included in the response body (in the "ProblemDetails" element).

On failure, the appropriate HTTP status code indicating the error shall be returned and appropriate additional error information should be returned in the POST response body.

5.3 Npanf_ResolveRemoteUserId Service

5.3.1 Service Description

This service enables an NF service consumer (i.e. SMF) to request the PANF to resolve the Remote User ID. The following are the key functionalities of this NF service.

- Resolve the Remote User ID

5.3.2 Service Operations

5.3.2.1 Introduction

5.3.2.2 ResolveRemoteUserId

5.3.2.2.1 General

The ResolveRemoteUserId service operation is invoked by a NF Service Consumer, i.e. SMF, towards the PANF to Resolve the Remote User ID.

The ResolveRemoteUserId service operation is used during the following procedure:

- PC5 security establishment for 5G ProSe UE-to-Network relay communication over Control Plane (see 3GPP TS 33.503 [14], clause 6.3.3.3.2)

The NF Service Consumer (i.e. SMF) shall resolve the Remote User ID by invoking the "get" custom method on the resource URI of "Resolve Remote User ID" resource, see clause 6.2.3.2.4. See also Figure 5.3.2.2.1-1.

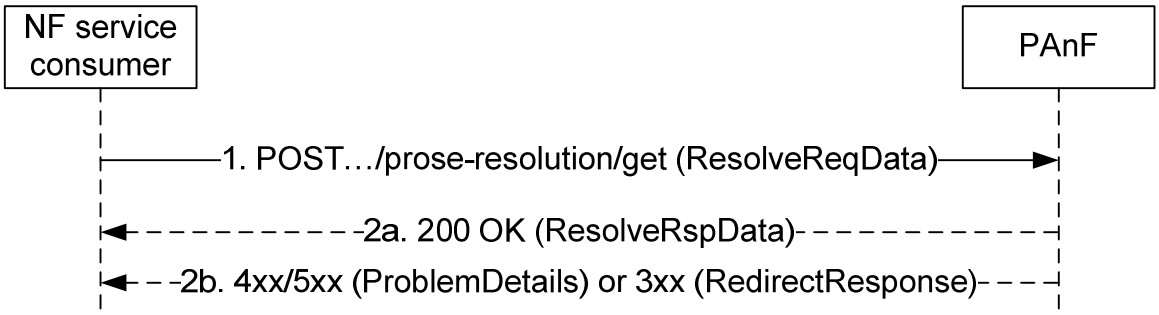

Figure 5.3.2.2-1: Resolve the Remote User ID

1. The NF service consumer sends a POST request to the resource representing the get custom operation. The request body shall contain the Remote User ID (CP-PRUK ID).

2a. On success, "200 OK" shall be returned. The response body shall contain the SUPI.

2b. On failure, the appropriate HTTP status code indicating the error shall be returned and appropriate additional error information should be returned.

6 API Definitions

6.1 Npanf_ProseKey Service API

6.1.1 Introduction

The Npanf_ProseKey service shall use the Npanf_ProseKey API.

The API URI of the Npanf_ProseKey API shall be:

{apiRoot}/<apiName>/<apiVersion>

The request URIs used in HTTP requests from the NF service consumer towards the NF service producer shall have the Resource URI structure defined in clause 4.4.1 of 3GPP TS 29.501 [5], i.e.:

{apiRoot}/<apiName>/<apiVersion>/<apiSpecificResourceUriPart>

with the following components:

- The {apiRoot} shall be set as described in 3GPP TS 29.501 [5].
- The <apiName> shall be "npanf-prosekey".
- The <apiVersion> shall be "v1".
- The <apiSpecificResourceUriPart> shall be set as described in clause 6.1.3.

6.1.2 Usage of HTTP

6.1.2.1 General

HTTP/2, IETF RFC 7540 [11], shall be used as specified in clause 5 of 3GPP TS 29.500 [4].

HTTP/2 shall be transported as specified in clause 5.3 of 3GPP TS 29.500 [4].

The OpenAPI [6] specification of HTTP messages and content bodies for the Npanf_ProseKey API is contained in Annex A.

6.1.2.2 HTTP standard headers

6.1.2.2.1 General

See clause 5.2.2 of 3GPP TS 29.500 [4] for the usage of HTTP standard headers.

6.1.2.2.2 Content type

JSON, IETF RFC 8259 [12], shall be used as content type of the HTTP bodies specified in the present specification as specified in clause 5.4 of 3GPP TS 29.500 [4]. The use of the JSON format shall be signalled by the content type "application/json".

"Problem Details" JSON object shall be used to indicate additional details of the error in a HTTP response body and shall be signalled by the content type "application/problem+json", as defined in IETF RFC 7807 [13].

6.1.2.3 HTTP custom headers

The mandatory HTTP custom header fields specified in clause 5.2.3.2 of 3GPP TS 29.500 [4] shall be supported, and the optional HTTP custom header fields specified in clause 5.2.3.3 of 3GPP TS 29.500 [4] may be supported.

6.1.3 Resources

6.1.3.1 Overview

This clause describes the structure for the Resource URIs and the resources and methods used for the service.

Figure 6.1.3.1-1 depicts the resource URIs structure for the Npanf_ProseKey API.

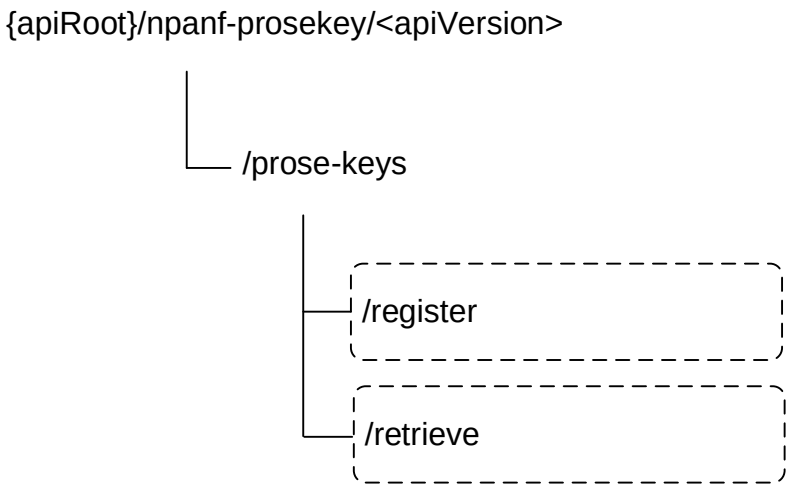

Figure 6.1.3.1-1: Resource URI structure of the Npanf_ProseKey API

Table 6.1.3.1-1 provides an overview of the resources and applicable HTTP methods.

Table 6.1.3.1-1: Resources and methods overview

Resource name	Resource URI	HTTP method or custom operation	Description
Prose Keys	/prose-keys/register	register (POST)	Store the Prose Context
	/prose-keys/retrieve	retrieve (POST)	Retrieve the CP-PRUK

6.1.3.2 Resource: Prose Keys

6.1.3.2.1 Description

This resource is used to represent Prose Keys Registration and Retrieval.

6.1.3.2.2 Resource Definition

Resource URI: {apiRoot}/npanf-prosekey/<apiVersion>/prose-keys

This resource shall support the resource URI variables defined in table 6.1.3.2.2-1.

Table 6.1.3.2.2-1: Resource URI variables for this resource

Name	Data type	Definition
apiRoot	string	See clause 6.1.1

6.1.3.2.3 Resource Standard Methods

No HTTP method has been defined for the Prose Keys resource.

6.1.3.2.4 Resource Custom Operations

6.1.3.2.4.1 Overview

Table 6.1.3.2.4.1-1: Custom operations

Operation name	Custom operation URI	Mapped HTTP method	Description
register	/prose-keys/register	POST	Store the Prose Context Info
retrieve	/prose-keys/retrieve	POST	Retrieve the CP-PRUK

6.1.3.2.4.2 Operation: register

6.1.3.2.4.2.1 Description

The registration custom operation is used by the NF service consumer (AUSF) to store the Prose context info. For details see 3GPP TS 33.503 [14].

6.1.3.2.4.2.2 Operation Definition

This operation shall support the request data structures specified in table 6.1.3.2.4.2.2-1 and the response data structure and response codes specified in table 6.1.3.2.4.2.2-2.

Table 6.1.3.2.4.2.2-1: Data structures supported by the POST Request Body on this resource

Data type	P	Cardinality	Description
ProseContextInfo	M	1	Contains the SUPI, CP-PRUK, CP-PRUK ID, RSC.

Table 6.1.3.2.4.2.2-2: Data structures supported by the POST Response Body on this resource

Data type	P	Cardinality	Response codes	Description
n/a			204 No Content	Upon success, an empty response body shall be returned.
ProblemDetails	O	0..1	404 Not Found	The "cause" attribute may be used to indicate one of the following application errors: - USER_NOT_FOUND
NOTE: The mandatory HTTP error status code for the POST method listed in Table 5.2.7.1-1 of 3GPP TS 29.500 [4] also apply.				

6.1.3.2.4.3 Operation: retrieve

6.1.3.2.4.3.1 Description

The retrieval custom operation is used by the NF service consumer (AUSF) to retrieve the Prose Key. For details see 3GPP TS 33.503 [14].

6.1.3.2.4.3.2 Operation Definition

This operation shall support the request data structures specified in table 6.1.3.2.4.3.2-1 and the response data structure and response codes specified in table 6.1.3.2.4.3.2-2.

Table 6.1.3.2.4.3.2-1: Data structures supported by the POST Request Body on this resource

Data type	P	Cardinality	Description
ProseKeyRequest	M	1	Contains the 5GPRUK ID, RSC.

Table 6.1.3.2.4.3.2-2: Data structures supported by the POST Response Body on this resource

Data type	P	Cardinality	Response codes	Description
ProseKeyResponse	M	1	200 OK	Upon success, a response body containing the Prose Key shall be returned.
ProblemDetails	O	0..1	404 Not Found	The "cause" attribute may be used to indicate one of the following application errors: - USER_NOT_FOUND - DATA_NOT_FOUND
NOTE: The mandatory HTTP error status code for the POST method listed in Table 5.2.7.1-1 of 3GPP TS 29.500 [4] also apply.				

6.1.4 Custom Operations without associated resources

There is no custom operation without associated resources supported in Npanf_ProseKey Service.

6.1.5 Notifications

There is no notification defined for Npanf_ProseKey service.

6.1.6 Data Model

6.1.6.1 General

This clause specifies the application data model supported by the API.

Table 6.1.6.1-1 specifies the data types defined for the Npanf_ProseKey service based interface protocol.

Table 6.1.6.1-1: Npanf_ProseKey specific Data Types

Data type	Clause defined	Description	Applicability
ProseContextInfo	6.1.6.2.2	ProSe Context Information	
ProseKeyRequest	6.1.6.2.3	Contains the CP-PRUK ID, RSC.	
ProseKeyResponse	6.1.6.2.4	Contains the CP-PRUK	
5GPruk	6.1.6.3.2	ProSe Remote User Key over Control Plane	

Table 6.1.6.1-2 specifies data types re-used by the Npanf_ProseKey service based interface protocol from other specifications, including a reference to their respective specifications and when needed, a short description of their use within the Npanf_ProseKey service based interface.

Table 6.1.6.1-2: Npanf_ProseKey re-used Data Types

Data type	Reference	Comments	Applicability
RelayServiceCode	3GPP TS 29.571 [15]	Relay Service Code	
Supi	3GPP TS 29.571 [15]	see 3GPP TS 23.501 [2] clause 5.9.2	
5GPrukId	3GPP TS 29.571 [15]	ProSe Remote User Key ID over Control Plane	

6.1.6.2 Structured data types

6.1.6.2.1 Introduction

This clause defines the structures to be used in resource representations.

6.1.6.2.2 Type: ProseContextInfo

Table 6.1.6.2.2-1: Definition of type ProseContextInfo

Attribute name	Data type	P	Cardinality	Description	Applicability
supi	Supi	M	1	The SUPI of the UE	
5gPrukId	5GPrukId	M	1	This IE shall indicate the CP-PRUK ID of the 5G ProSe Remote UE.	
5gPruk	5GPruk	M	1	This IE shall indicate the CP-PRUK of the 5G ProSe Remote UE.	
relayServiceCode	RelayServiceCode	M	1	This IE shall indicate the Relay Service Code from the 5G ProSe Remote UE.	

6.1.6.2.3 Type: ProseKeyRequest

Table 6.1.6.2.3-1: Definition of type ProseKeyRequest

Attribute name	Data type	P	Cardinality	Description	Applicability
5gPrukId	5GPrukId	M	1	This IE shall indicate the CP-PRUK ID of the 5G ProSe Remote UE.	
relayServiceCode	RelayServiceCode	M	1	This IE shall indicate the Relay Service Code from the 5G ProSe Remote UE.	

6.1.6.2.4 Type: ProseKeyResponse

Table 6.1.6.2.4-1: Definition of type ProseKeyResponse

Attribute name	Data type	P	Cardinality	Description	Applicability
5gPruk	5GPruk	M	1	This IE shall indicate the CP-PRUK of the 5G ProSe Remote UE.	

6.1.6.3 Simple data types and enumerations

6.1.6.3.1 Introduction

This clause defines simple data types and enumerations that can be referenced from data structures defined in the previous clauses.

6.1.6.3.2 Simple data types

The simple data types defined in table 6.1.6.3.2-1 shall be supported.

Table 6.1.6.3.2-1: Simple data types

Type Name	Type Definition	Description	Applicability
5GPruk	string	ProSe Remote User Key over Control Plane String value carrying the CP-PRUK in hexadecimal presentation. pattern: "^[A-Fa-f0-9]{64}$"	

6.1.6.4 Data types describing alternative data types or combinations of data types

There is no data type describing alternative data types or combinations of data types in Npanf_ProseKey Service.

6.1.6.5 Binary data

There is no binary data type in Npanf_ProseKey Service.

6.1.7 Error Handling

6.1.7.1 General

For the Npanf_ProseKey API, HTTP error responses shall be supported as specified in clause 4.8 of 3GPP TS 29.501 [5]. Protocol errors and application errors specified in table 5.2.7.2-1 of 3GPP TS 29.500 [4] shall be supported for an HTTP method if the corresponding HTTP status codes are specified as mandatory for that HTTP method in table 5.2.7.1-1 of 3GPP TS 29.500 [4].

In addition, the requirements in the following clauses are applicable for the Npanf_ProseKey API.

6.1.7.2 Protocol Errors

No specific procedures for the Npanf_ProseKey service are specified.

6.1.7.3 Application Errors

The application errors defined for the Npanf_ProseKey service are listed in Table 6.1.7.3-1.

Table 6.1.7.3-1: Application errors

Application Error	HTTP status code	Description
USER_NOT_FOUND	404 Not Found	The provided subscriber identifier is not found.
DATA_NOT_FOUND	404 Not Found	The requested data is not found/does not exist.

6.1.8 Feature negotiation

The optional features in table 6.1.8-1 are defined for the Npanf_ProseKey API. They shall be negotiated using the extensibility mechanism defined in clause 6.6 of 3GPP TS 29.500 [4].

Table 6.1.8-1: Supported Features

Feature number	Feature Name	Description

6.1.9 Security

As indicated in 3GPP TS 33.501 [8] and 3GPP TS 29.500 [4], the access to the Npanf_ProseKey API may be authorized by means of the OAuth2 protocol (see IETF RFC 6749 [9]), based on local configuration, using the "Client Credentials" authorization grant, where the NRF (see 3GPP TS 29.510 [10]) plays the role of the authorization server.

If OAuth2 is used, an NF Service Consumer, prior to consuming services offered by the Npanf_ProseKey API, shall obtain a "token" from the authorization server, by invoking the Access Token Request service, as described in 3GPP TS 29.510 [10], clause 5.4.2.2.

NOTE: When multiple NRFs are deployed in a network, the NRF used as authorization server is the same NRF that the NF Service Consumer used for discovering the Npanf_ProseKey service.

The Npanf_ProseKey API defines a single scope "npanf-prosekey" for the entire service, and it does not define any additional scopes at resource or operation level.

6.2 Npanf_ResolveRemoteUserId Service API

6.2.1 Introduction

The Npanf_ResolveRemoteUserId shall use the Npanf_ResolveRemoteUserId API.

The API URI of the Npanf_ResolveRemoteUserId API shall be:

{apiRoot}/<apiName>/<apiVersion>

The request URIs used in HTTP requests from the NF service consumer towards the NF service producer shall have the Resource URI structure defined in clause 4.4.1 of 3GPP TS 29.501 [6], i.e.:

{apiRoot}/<apiName>/<apiVersion>/<apiSpecificResourceUriPart>

with the following components:

- The {apiRoot} shall be set as described in 3GPP TS 29.501 [6].
- The <apiName> shall be "npanf-userid".
- The <apiVersion> shall be "v1".
- The <apiSpecificResourceUriPart> shall be set as described in clause 6.2.3.

6.2.2 Usage of HTTP

6.2.2.1 General

HTTP/2, IETF RFC 7540 [8], shall be used as specified in clause 5 of 3GPP TS 29.500 [5].

HTTP/2 shall be transported as specified in clause 5.3 of 3GPP TS 29.500 [5].

The OpenAPI [7] specification of HTTP messages and content bodies for the Npanf_ResolveRemoteUserId API is contained in Annex A.

6.2.2.2 HTTP standard headers

6.2.2.2.1 General

See clause 5.2.2 of 3GPP TS 29.500 [5] for the usage of HTTP standard headers.

6.2.2.2.2 Content type

JSON, IETF RFC 8259 [9], shall be used as content type of the HTTP bodies specified in the present specification as specified in clause 5.4 of 3GPP TS 29.500 [5]. The use of the JSON format shall be signalled by the content type "application/json".

"Problem Details" JSON object shall be used to indicate additional details of the error in a HTTP response body and shall be signalled by the content type "application/problem+json", as defined in IETF RFC 7807 [10].

6.2.2.3 HTTP custom headers

The mandatory HTTP custom header fields specified in clause 5.2.3.2 of 3GPP TS 29.500 [4] shall be supported, and the optional HTTP custom header fields specified in clause 5.2.3.3 of 3GPP TS 29.500 [4] may be supported.

6.2.3 Resources

6.2.3.1 Overview

This clause describes the structure for the Resource URIs and the resources and methods used for the service.

Figure 6.2.3.1-1 describes the resource URI structure of the Npanf_ResolveRemoteUserId API.

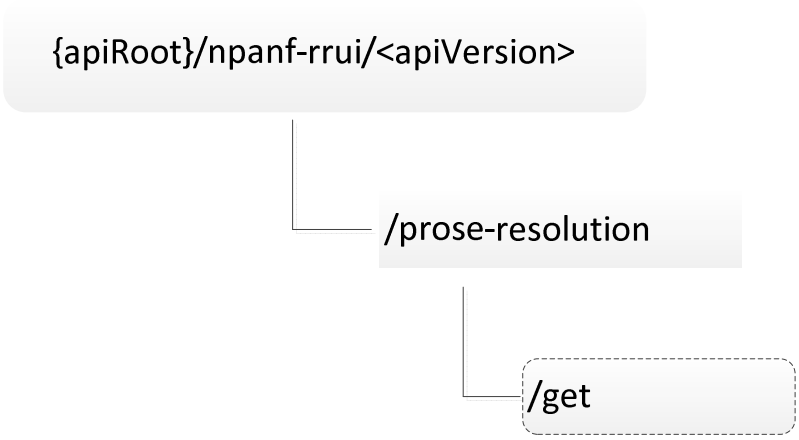

Figure 6.2.3.1-1: Resource URI structure of the Npanf_ResolveRemoteUserId API

Table 6.2.3.1-1 provides an overview of the resources and applicable HTTP methods.

Table 6.2.3.1-1: Resources and methods overview

Resource name	Resource URI	HTTP method or custom operation	Description
Resolve Remote User ID	/prose-resolution	get (POST)	ResolveRemoteUserId service operation

6.2.3.2 Resource: Resolve Remote User ID

6.2.3.2.1 Description

This resource represents resolving Remote User ID by the PAnF.

6.2.3.2.2 Resource Definition

Resource URI: **{apiRoot}/<apiName>/<apiVersion>/prose-resolution**

This resource shall support the resource URI variables defined in table 6.2.3.2.2-1.

Table 6.2.3.2.2-1: Resource URI variables for this resource

Name	Data type	Definition
apiRoot	string	See clause 6.2.1

6.2.3.2.3 Resource Standard Methods

There is no standard method supported by the resource.

6.2.3.2.4 Resource Custom Operations

6.2.3.2.4.1 Overview

Table 6.2.3.2.4.1-1: Custom operations

Operation name	Custom operaration URI	Mapped HTTP method	Description
get	{resourceUri}/get	POST	ResolveRemoteUserId service operation

6.2.3.2.4.2 Operation: get

6.2.3.2.4.2.1 Description

This custom operation requests to resolve the Remote User ID by the PAnF.

6.2.3.2.4.2.2 Operation Definition

This operation shall support the request data structures specified in table 6.2.3.2.4.2.2-1 and the response data structure and response codes specified in table 6.2.3.2.4.2.2-2.

Table 6.2.3.2.4.2.2-1: Data structures supported by the POST Request Body on this resource

Data type	P	Cardinality	Description
ResolveReqData	M	1	Resolve Request Data, including the Remote User ID of the 5G ProSe Remote UE.

Table 6.2.3.2.4.2.2-2: Data structures supported by the POST Response Body on this resource

Data type	P	Cardinality	Response codes	Description
ResolveRspData	M	1	200 OK	Resolve Response Data, including the SUPI of the 5G ProSe Remote UE.
ProblemDetails	O	0..1	404 Not Found	The "cause" attribute shall be set to one of the following application error: - USER_NOT_FOUND See table 6.2.7.3-1 for the description of these errors.
NOTE1: The manadatory HTTP error status code for the POST method listed in Table 5.2.7.1-1 of 3GPP TS 29.500 [5] also apply. NOTE 2: RedirectResponse may be inserted by an SCP, see clause 6.10.9.1 of 3GPP TS 29.500 [4].				

6.2.4 Custom Operations without associated resources

There is no custom operation without associated resources supported in Npanf_ResolveRemoteUserId Service.

6.2.5 Notifications

There is no notification defined for Npanf_ResolveRemoteUserId service.

6.2.6 Data Model

6.2.6.1 General

This clause specifies the application data model supported by the API.

Table 6.2.6.1-1 specifies the data types defined for the Npanf_ResolveRemoteUserId service based interface protocol.

Table 6.2.6.1-1: Npanf_ResolveRemoteUserId specific Data Types

Data type	Clause defined	Description	Applicability
ResolveReqData	6.2.6.2.2	Resolve Request Data, including the Remote User ID of the 5G ProSe Remote UE.	
ResolveRspData	6.2.6.2.3	Resolve Response Data, including the SUPI of the 5G ProSe Remote UE.	

Table 6.2.6.1-2 specifies data types re-used by the Npanf_ResolveRemoteUserId service based interface protocol from other specifications, including a reference to their respective specifications and when needed, a short description of their use within the Npanf_ResolveRemoteUserId service based interface.

Table 6.2.6.1-2: Npanf_ResolveRemoteUserId re-used Data Types

Data type	Reference	Comments	Applicability
Supi	3GPP TS 29.571 [15]	Supi	
5GPrukId	3GPP TS 29.571 [15]	ProSe Remote User Key ID over Control Plane	

6.2.6.2 Structured data types

6.2.6.2.1 Introduction

This clause defines the structures to be used in resource representations.

6.2.6.2.2 Type: ResolveReqData

Table 6.2.6.2.2-1: Definition of type ResolveReqData

Attribute name	Data type	P	Cardinality	Description	Applicability
cpPrukId	5GPrukId	M	1	This IE shall indicate the CP-PRUK ID of the 5G ProSe Remote UE.	

6.2.6.2.3 Type: ResolveRspData

Table 6.2.6.2.3-1: Definition of type ResolveRspData

Attribute name	Data type	P	Cardinality	Description	Applicability
supi	Supi	M	1	This IE shall indicate the SUPI.	

6.2.6.3 Simple data types and enumerations

6.2.6.3.1 Introduction

This clause defines simple data types and enumerations that can be referenced from data structures defined in the previous clauses.

6.2.6.3.2 Simple data types

The simple data types defined in table 6.2.6.3.2-1 shall be supported.

Table 6.2.6.3.2-1: Simple data types

Type Name	Type Definition	Description	Applicability

6.2.6.4 Data types describing alternative data types or combinations of data types

There is no data type describing alternative data types or combinations of data types in Npanf_ResolveRemoteUserId Service.

6.2.6.5 Binary data

There is no binary data type in Npanf_ResolveRemoteUserId Service.

6.2.7 Error Handling

6.2.7.1 General

For the Npanf_ResolveRemoteUserId API, HTTP error responses shall be supported as specified in clause 4.8 of 3GPP TS 29.501 [6]. Protocol errors and application errors specified in table 5.2.7.2-1 of 3GPP TS 29.500 [5] shall be supported for an HTTP method if the corresponding HTTP status codes are specified as mandatory for that HTTP method in table 5.2.7.1-1 of 3GPP TS 29.500 [5].

In addition, the requirements in the following clauses are applicable for the Npanf_ResolveRemoteUserId API.

6.2.7.2 Protocol Errors

Protocol errors handling shall be supported as specified in clause 5.2.7 of 3GPP TS 29.500 [5].

6.2.7.3 Application Errors

The application errors defined for the Npanf_ResolveRemoteUserId service are listed in Table 6.2.7.3-1.

Table 6.2.7.3-1: Application errors

Application Error	HTTP status code	Description
USER_NOT_FOUND	404 Not Found	The provided subscriber identifier is not found.

6.2.8 Feature negotiation

The optional features in table 6.2.8-1 are defined for the Npanf_ResolveRemoteUserId API. They shall be negotiated using the extensibility mechanism defined in clause 6.6 of 3GPP TS 29.500 [5].

Table 6.2.8-1: Supported Features

Feature number	Feature Name	Description
N/A		

6.2.9 Security

As indicated in 3GPP TS 33.501 [11] and 3GPP TS 29.500 [5], the access to the Npanf_ResolveRemoteUserId API may be authorized by means of the OAuth2 protocol (see IETF RFC 6749 [12]), based on local configuration, using the

"Client Credentials" authorization grant, where the NRF (see 3GPP TS 29.510 [13]) plays the role of the authorization server.

If OAuth2 is used, an NF Service Consumer, prior to consuming services offered by the Npanf_ResolveRemoteUserId API, shall obtain a "token" from the authorization server, by invoking the Access Token Request service, as described in 3GPP TS 29.510 [13], clause 5.4.2.2.

NOTE: When multiple NRFs are deployed in a network, the NRF used as authorization server is the same NRF that the NF Service Consumer used for discovering the Npanf_ResolveRemoteUserId service.

The Npanf_ResolveRemoteUserId API defines a single scope "npanf-userid" for OAuth2 authorization (as specified in 3GPP TS 33.501 [11]) for the entire service, and it does not define any additional scopes at resource or operation level.

Annex A (normative): OpenAPI specification

A.1 General

This Annex specifies the formal definition of the API(s) defined in the present specification. It consists of OpenAPI specifications in YAML format.

This Annex takes precedence when being discrepant to other parts of the specification with respect to the encoding of information elements and methods within the API(s).

NOTE 1: The semantics and procedures, as well as conditions, e.g. for the applicability and allowed combinations of attributes or values, not expressed in the OpenAPI definitions but defined in other parts of the specification also apply.

Informative copies of the OpenAPI specification files contained in this 3GPP Technical Specification are available on a Git-based repository that uses the GitLab software version control system (see clause 5.3.1 of 3GPP TS 29.501 [5] and clause 5B of 3GPP TR 21.900 [7]).

A.2 Npanf_ProseKey API

```
openapi: 3.0.0

info:
  title: Npanf_ProseKey
  version: 1.0.1
  description: |
    PAnF ProseKey Service.
    © 2022, 3GPP Organizational Partners (ARIB, ATIS, CCSA, ETSI, TSDSI, TTA, TTC).
    All rights reserved.

externalDocs:
  description: 3GPP TS 29.553 V17.1.0; 5G System; 5G ProSe Anchor Services; Stage 3.
  url: https://www.3gpp.org/ftp/Specs/archive/29_series/29.553/

servers:
  - url: '{apiRoot}/npanf-prosekey/<apiVersion>'
    variables:
      apiRoot:
        default: https://example.com
        description: apiRoot as defined in clause 4.4 of 3GPP TS 29.501

security:
  - {}
  - oAuth2ClientCredentials:
    - npanf-prosekey

paths:
  /prose-keys/register:
    post:
      summary: Register the Prose Key
      operationId: ProseKeyRegistration
      tags:
        - prosekey registration
      requestBody:
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/ProseContextInfo'
        required: true
      responses:
        '204':
          description: Successful Response
        '404':
          $ref: 'TS29571_CommonData.yaml#/components/responses/404'
        default:
          description: Unexpected error
```

```
  /prose-keys/retrieve:
    post:
      summary: retrieve the prose key
      operationId: ProseKeyRetrieval
      tags:
        - Prose Key Retrieval
      requestBody:
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/ProseKeyRequest'
        required: true
      responses:
        '200':
          description: Expected response to a valid request
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/ProseKeyResponse'
        '400':
          $ref: 'TS29571_CommonData.yaml#/components/responses/400'
        '404':
          $ref: 'TS29571_CommonData.yaml#/components/responses/404'
        '500':
          $ref: 'TS29571_CommonData.yaml#/components/responses/500'
        '503':
          $ref: 'TS29571_CommonData.yaml#/components/responses/503'
        default:
          description: Unexpected error

components:
  securitySchemes:
    oAuth2ClientCredentials:
      type: oauth2
      flows:
        clientCredentials:
          tokenUrl: '{nrfApiRoot}/oauth2/token'
          scopes:
            npanf-prosekey: Access to the Npanf_ProseKey API

  schemas:
#
#  Structured Data Types
#
    ProseContextInfo:
      description: Prose Context Info.
      type: object
      properties:
        supi:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Supi'
        5gPruk:
          $ref: '#/components/schemas/5GPruk'
        5gPrukId:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/5GPrukId'
        relayServiceCode:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/RelayServiceCode'
      required:
        - supi
        - 5gPruk
        - 5gPrukId
        - relayServiceCode

    ProseKeyRequest:
      description: Prose Key Request.
      type: object
      properties:
        5gPrukId:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/5GPrukId'
        relayServiceCode:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/RelayServiceCode'
      required:
        - 5gPrukId
        - relayServiceCode

    ProseKeyResponse:
      description: Prose Key Response.
      type: object
```

```
      properties:
        5gPruk:
          $ref: '#/components/schemas/5GPruk'
      required:
        - 5gPruk

#
#  Simple Data Types
#
    5GPruk:
      description: ProSe Remote User Key over Control Plane
      type: string
      pattern: '^[A-Fa-f0-9]{64}$'

#
#  Enumeration Data Types
#
```

A.3 Npanf_ResolveRemoteUserId API

```
openapi: 3.0.0
info:
  version: '1.0.0'
  title: 'Npanf_ResolveRemoteUserId'
  description: |
    PANF Resolve Remote User Id Service.
    © 2023, 3GPP Organizational Partners (ARIB, ATIS, CCSA, ETSI, TSDSI, TTA, TTC).
    All rights reserved.

externalDocs:
  description: 3GPP TS 29.559 V17.2.0; 5G System; 5G ProSe Key Management Services; Stage 3.
  url: https://www.3gpp.org/ftp/Specs/archive/29_series/29.559/

servers:
  - url: '{apiRoot}/npanf-userid/v1'
    variables:
      apiRoot:
        default: https://example.com
        description: apiRoot as defined in clause 4.4 of 3GPP TS 29.501

security:
  - {}
  - oAuth2ClientCredentials:
    - npanf-userid

paths:
  /prose-resolution/get:
    post:
      summary: Request to resolve the Remote User ID
      operationId: ProseResolve
      tags:
        - Resolve Remote User ID
      requestBody:
        required: true
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/ResolveReqData'
      responses:
        '200':
          description: Success
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/ResolveRspData'
        '400':
          $ref: 'TS29571_CommonData.yaml#/components/responses/400'
        '401':
          $ref: 'TS29571_CommonData.yaml#/components/responses/401'
        '403':
          $ref: 'TS29571_CommonData.yaml#/components/responses/403'
        '404':
          $ref: 'TS29571_CommonData.yaml#/components/responses/404'
        '411':
```

```
          $ref: 'TS29571_CommonData.yaml#/components/responses/411'
        '413':
          $ref: 'TS29571_CommonData.yaml#/components/responses/413'
        '415':
          $ref: 'TS29571_CommonData.yaml#/components/responses/415'
        '429':
          $ref: 'TS29571_CommonData.yaml#/components/responses/429'
        '500':
          $ref: 'TS29571_CommonData.yaml#/components/responses/500'
        '502':
          $ref: 'TS29571_CommonData.yaml#/components/responses/502'
        '503':
          $ref: 'TS29571_CommonData.yaml#/components/responses/503'
        default:
          description: Unexpected error

components:
  securitySchemes:
    oAuth2ClientCredentials:
      type: oauth2
      flows:
        clientCredentials:
          tokenUrl: '{nrfApiRoot}/oauth2/token'
          scopes:
            npanf-userid: Access to the Npanf_ResolveRemoteUserId API

  schemas:
#
#  Structured Data Types
#
    ResolveReqData:
      description: Representation of the Resolve Request Data.
      type: object
      properties:
        cpPrukId:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/5GPrukId'
      required:
        - cpPrukId

    ResolveRspData:
      description: Representation of the Resolve Response Data.
      type: object
      properties:
        supi:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Supi'
      required:
        - supi

#
#  Simple Data Types
#


#
#  Enumeration Data Types
#
```

Annex B (informative): Change history

Change history							
Date	**Meeting**	**TDoc**	**CR**	**Rev**	**Cat**	**Subject/Comment**	**New version**
2022-08	CT4#111 e	C4-224498				Initial draft Implementation of pCRs agreed in CT4#111-e including C4-224430,C4-224046,C4-224431,C4-224432,C4-224592,C4-224433,C4-224593,C4-224373,C4-224374,C4-224594,C4-224595,C4-224377,C4-224378,C4-224596	0.1.0
2022-09	CT#97-e	CP-222239				TS presented for information and approval	1.0.0
2022-09	CT#97-e					Approved in CT#97-e	17.0.0
2022-12	CT#98-e	CP-223054	0002	2	F	5GPRUK Name Alignment	17.1.0
2022-12	CT#98-e	CP-223054	0003	1	F	Remove editor's note and white spaces	17.1.0
2022-12	CT#98-e	CP-223066	0008	-	F	29.553 Rel-17 API version and External doc update	17.1.0
2023-06	CT#100	CP-231078	0014	1	F	Adding Npanf_ResolveRemoteUserId service	17.2.0
2023-06	CT#100	CP-231085	0019	-	F	29.553 Rel-17 API version and External doc update	17.2.0

History

Document history		
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